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14. O_2^{2-} is diamagnetic. Additional two electrons paired in antibonding molecular orbital $\pi^*_{2P_y}$ and $\pi^*_{2P_z}$ (option (b))

15. Bond order = $\frac{1}{2}(n_b - n_a)$

$2.5 = \frac{1}{2}(8 - n_a)$

$\Rightarrow 5 = 8 - n_a$

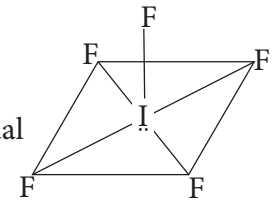
$\Rightarrow n_a = 8\text{-}5 = 3$ (option a)

16. IF_5 - 5 bond pair + 1 lone pair

hybridisation Sp^3d^2

Shape

Square pyramidal



17. Correct statement :

All five Sp^3d hybrid orbitals are equivalent

$\therefore$ incorrect statement : Option (c)

18. SeF_4, XeO_2F_2 - Sp^3d hybridisation,

option (a) T-shaped, one line pair on central atom.

19. H_2O - Central atom Sp^3 hybridised

NO_2^- - Central atom Sp^2 hybridised

BF_3 - Central atom Sp^2 hybridised

NH_2^- - Central atom Sp^3 hybridised

option (c) is correct.

20. NO_3^- - Sp^2 hybridisation, planar

H_3O^+ - Sp^3 hybridisation, pyramidal

option (a) is correct.

21. $\overset{Sp^3}{CH_3} - \overset{Sp^2}{CH} = \overset{Sp}{C} = \overset{Sp^3}{CH} - \overset{Sp^3}{CH_3}$ Option (a)

22. XeF_2 is isostructural with ICl_2^-

23. $\overset{Sp^3}{CH_4}$, $\overset{Sp^3}{CH_3} - CH_3$, $\overset{Sp^2}{CH_2} = CH_2$, $\overset{Sp}{CH} \equiv CH$

25, 25, 33.3, 50

option (a) is correct.

24. Co_2 - Linear

C_2H_2 - Linar

option (c) is correct.

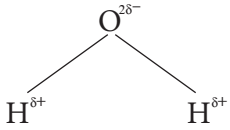
25. l.p - l.p > l.p - b.p > b.p - b.p

option (c) is correct

26. ClF_3 - Sp^3d hybridisation, 'T' shaped

option (c) is correct.

27.



option (d)

28. Correct statement is - the resonance hybrid should have lower energy than any of the contributing structure

Hence is correct statement is option (c)

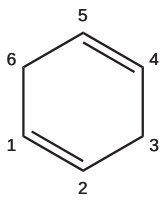
29. NH_4Cl (option (a))

30. 4U

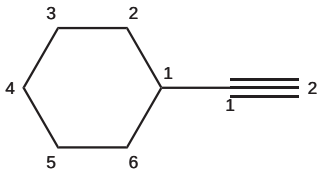
UNIT - 11

EVALUATE YOURSELF - KEY

2.

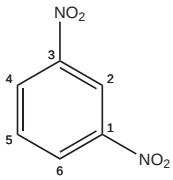


cyclohexa-1,4-diene

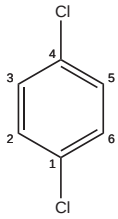


ethynylcyclohexane

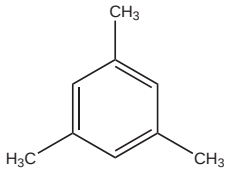
3.



1,3-dinitrobenzene



p-dichlorobenzene



1,3,5-trimethylbenzene